higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1: COLD-WATER RETICULATION

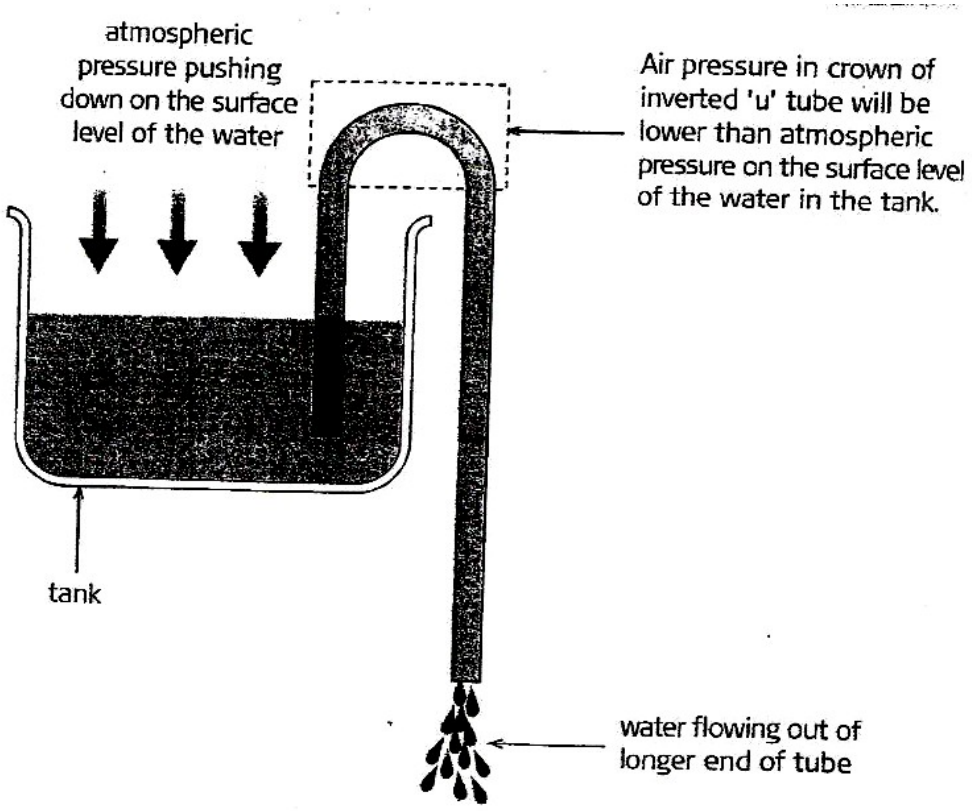
1.1 To read/regulate the amount of water used (1)

1.2 After the municipal stopcock and in front of the consumer's stopcock (1)

1.3 It is the water pressure in the system when no water is flowing. (2)

1.4
- Underground soil conditions that may affect pipework
- Unexpected obstacles like large rocks
- Underground root systems of trees in the area
- Paving or road surfacing where the pipes must be laid
- Pipe ducts in buildings
- Existing concrete floors and slabs
- Different types of external walls
- Roof trusses in the way
- Historical buildings where installations must match existing pipework
- Limited ceiling space (Any 4 × 1) (4)

1.5



(6 for sketch + 4 for labelling) (10)

1.6 To stop or prevent solids or other strange objects from entering the water system and cause blockages or pollute the components downstream (2)
[20]

QUESTION 2: HOT-WATER RETICULATION

2.1
- The water pressure is quite low (because of the height limit).
- The water flow can be slow because of the low pressure.
- There is very little reserve water in the tank.
- Modern mixers cannot be used as water pressure is too low.
- Airlocks occur in hot-water pipes. (Any 2 × 1) (2)

2.2
1 Cold-water supply
2 Stopcock
3 Open-swivel outlet
4 Outer casing
5 Insulating material
6 Hot-water cylinder
7 Hot-water feed pipe (overflow)
8 Temperature and pressure-safety valve
9 Vacuum breaker
10 Antisiphonage loop

(10 × 1) (10)

2.3
2.3.1 Yellow
2.3.2 Orange
2.3.3 Blue
2.3.4 Black
2.3.5 Red

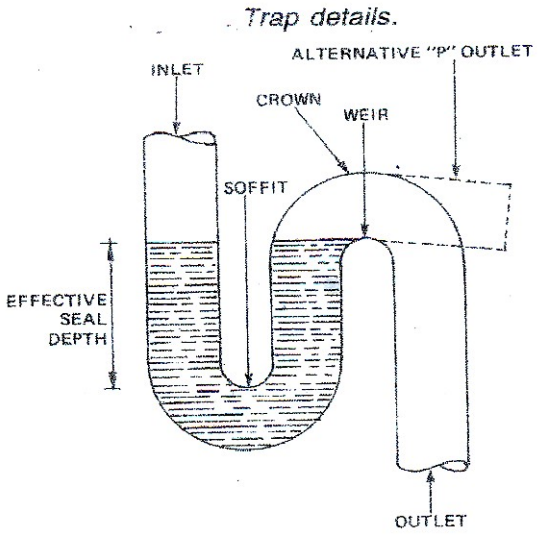
(5 × 1) (5)

2.4 When the water is heated and the system is not in use, the pressure-relief valve will allow a little water to escape to prevent the pressure inside the system from exceeding the present maximum pressure rating. (3)

[20]

QUESTION 3: ABOVE-GROUND DRAINAGE SYSTEM

3.1



(6)

3.2
- Close-coupled suite
- Low-level suite
- High-level suite (3)

3.3 It is the principle of connecting the individual water seal traps from a sanitary fixture to a ventilated drain or other ventilated pipe which leads into the open air. (3)

3.4
- Flushes away all soil water
- Has no blockage
- Traps and prevents foul or bad smelling air from the drain to enter the building
- Allows foul air and gasses building up in the drain to escape through the pipework into the atmosphere (air) (Any 3 × 1) (3)

[15]

QUESTION 4: SANITARY INSTALLATION

4.1
4.1.1 Access opening
4.1.2 Bidet
4.1.3 Cleaning eye
4.1.4 Inspection eye
4.1.5 Wash hand basin

(5 × 1) (5)

4.2
- Knock out the required holes for the taps or mixer.
- Fit the taps or mixer to the basin. Remember when you face the basin the cold water tap should be fitted on the right hand side and the hot water tap on the left.
- Tip the basin over onto an old carpet or a piece of soft cardboard. The surface will protect the basin from scratches or other marks.
- Fit and tighten the washers and back nuts to the taps underneath the basin. (4)

4.3 The water-closet (toilet) pan has a built-in water seal trap and will therefore not need to be fitted with a water seal trap. (1)

4.4
4.4.1 Bath/Wash hand basin
4.4.2 Wash hand basin
4.4.3 Cisterns/Storage tanks
4.4.4 Cisterns/Storage tanks
4.4.5 Ring main pipes (in water reticulation)

(5 × 1) (5)

[15]

QUESTION 5: RAINWATER SYSTEM

5.1 Fit the first bracket to the one side of the fascia as high as possible. Measure the length of the gutter and calculate its vertical fall. Fit the last bracket in position on the other end of the fascia. Stretch a fish line or the plumb bob line between these two brackets. Calculate the number of brackets that should go between the ends. The maximum distance between the brackets is 1 meter. Mark the position of all the brackets and fix them to the fascia using the line as height gauge. Remove the line. (8)

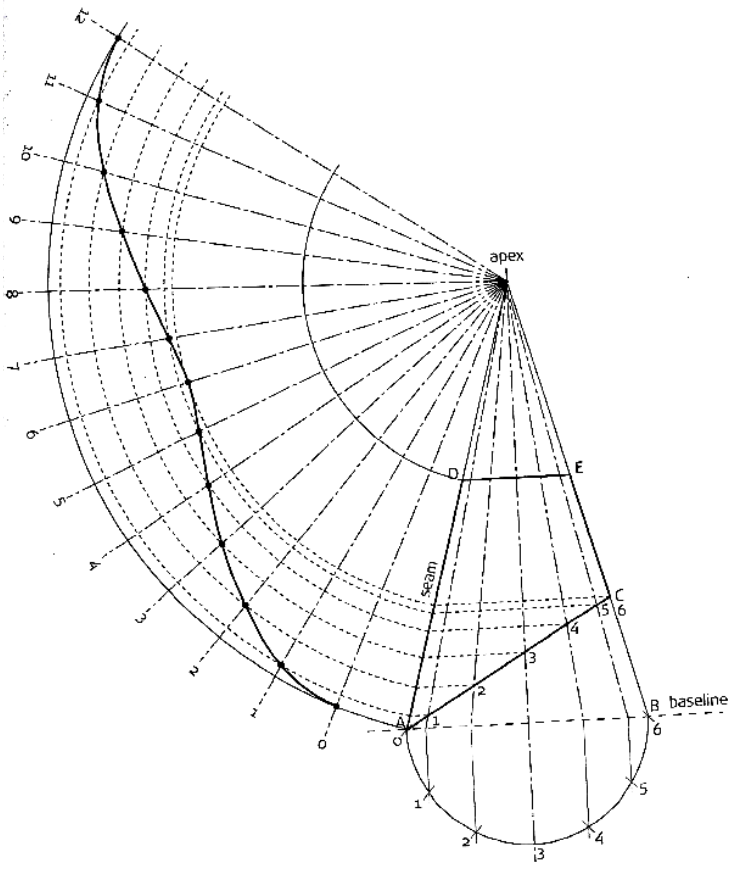
5.2 A rainwater system leads rainwater down from the roof and away from the house. (2)

5.3
- Unplasticised polyvinyl chloride (uPVC)
- Fibre cement
- Galvanised mild steel (GMS)
- Aluminium (Any 3 × 1) (3)

5.4
- Soldering method
- Pop-riveting method
- Groover method (Any 2 × 1) (2)

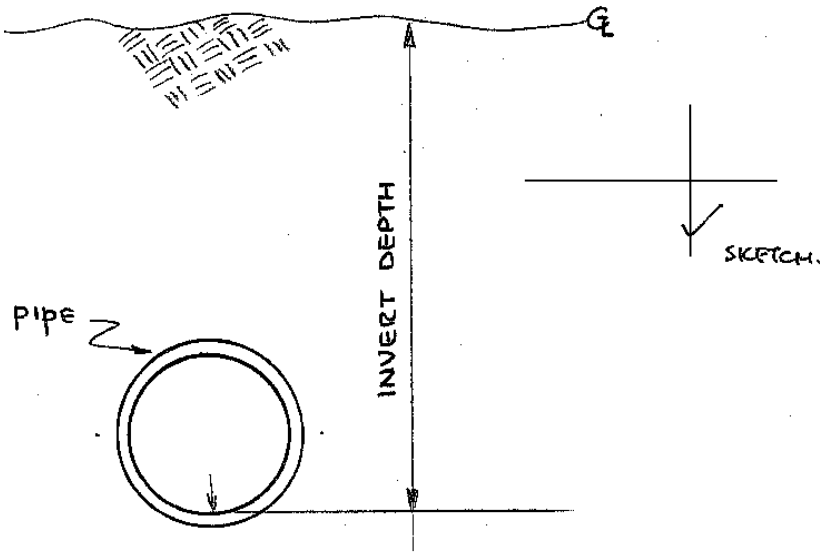
[15]

QUESTION 6: VENT-PIPE FLUSHING



(6 for pattern + 4 for cone) **[10]**

QUESTION 7: TRENCHING, EXCAVATION AND TRENCH SUPPORT



(3 for drawing + 2 for label) **[5]**

TOTAL: 100

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